

Turtle Drawing Cheatsheet

Quick commands for the turtle drawing template. Edit `src/turtle_drawing/drawings.py`, then run `uv run interactive`, `uv run live`, or `uv run render`.

<code>uv sync</code>	Install the project dependencies.
<code>uv run interactive</code>	Open the real-time Turtle window. Reloads when you save.
<code>uv run live</code>	Open the auto-refresh PNG preview window.
<code>uv run render</code>	Create output/drawing.png once.
<code>uv run watch</code>	Regenerate output/drawing.png whenever code changes.

Move

```
t.forward(100)
t.backward(50)
t.left(90)
t.right(45)
t.goto(10, 20)
t.setheading(0)
t.home()
```

Pen

```
t.penup()
t.pendown()
t.pensize(5)
t.color("red")
t.pencolor("blue")
current = t.pencolor()
```

Shapes

```
draw_square(t, 80)
draw_rectangle(t, 120, 60)
draw_triangle(t, 90)
draw_circle(t, 35)
draw_star(t, 70)
draw_flower(t, x, y)
```

Fill

```
t.fillcolor("yellow")
t.begin_fill()
draw_square(t, 80)
t.end_fill()
draw_filled_circle(t, 25, "red", "pink")
```

Text, Dots, Images

```
t.write("Hello")
t.dot(20, "green")
import_picture(t, "assets/example_picture.png", 0,
0, width=120)
pixel = get_pixel_color(t, 0, 0)
```

Jump Without Drawing

```
jump_to(t, -100, 50)
def jump_to(t, x, y):
    t.penup()
    t.goto(x, y)
    t.pendown()
```

Interactive Mode

<code>def on_click(t, x, y):</code>	Runs when the user clicks the Turtle window.
<code>def on_key(t, key):</code>	Runs when the user presses a supported key.
<code>r / g / b</code>	Change the current pen color in the example.
arrow keys	Move the turtle and draw in the example.
space	Draw a star in the example.
<code>t.clear()</code>	Clear the window.
<code>draw_picture(t)</code>	Draw the starting picture again.
<code>t.hideturtle()</code>	Hide the turtle cursor while drawing.
<code>t.speed(1)</code>	Slow drawing down. Use 0 for fastest.

Turtle coordinates

(0, 0) is the center. x goes left/right. y goes down/up. Use `jump_to(t, x, y)` when you want to move without drawing a line.